

VPA IB DP CS — Year 2 Semester 1 Overview

VPA IB DP Computer Science — Year 2 Semester 1 Overview

HL Topics: 5 (Abstract Data Structures) + 6 (Resource Management) | SL Focus: Revision + IA Completion

Semester Overview

Year 2 Semester 1 splits the cohort: HL students study Topics 5 and 6 (Abstract Data Structures and Resource Management), while SL students consolidate Year 1 content and complete their IA. This is the most demanding semester for both levels.

Pathways

Level	Weeks 1 – 9	Weeks 10 – 18
HL	Topic 5: Abstract Data Structures	Topic 6: Resource Management
SL	Revision (Topics 1 – 4) + IA Development	Revision (Topics 1 – 4) + IA Completion

IA Timeline (All Students)

Week	Milestone	Criterion
1	Final client meeting; success criteria locked	A
3	Record of tasks complete; design document submitted	A + B
6	Core functionality complete; video evidence planned	C
9	Full system coded; testing begins	C + D
12	Testing complete; evaluation draft	D + E
15	First full draft submitted for feedback	A – E
17	Final submission	A – E

Assessment Schedule

Assessment	Week	Who	Format
Topic 5 Progress Check	5	HL only	Data structure implementation (30 marks)
Topic 5 End-of-Topic Test	9	HL only	Paper 1 style (40 marks)
SL Revision Test 1	5	SL only	Topics 1 – 2 (40 marks)

Assessment	Week	Who	Format
SL Revision Test 2	9	SL only	Topics 3–4 (40 marks)
IA First Draft	15	All	Full Criteria A–E draft
Topic 6 Progress Check	14	HL only	OS concepts + calculations (25 marks)
Semester 1 Examination	18	All	Paper 1 mock (SL 1h30 / HL 2h10)

TOK This Semester

Topic 5: Abstract data structures like trees and graphs are mathematical abstractions. Do they exist independently of their computer implementations? Is a "tree" in code the same kind of thing as a tree in nature—or is it merely a useful fiction?

Topic 6: Operating systems allocate resources "fairly" using algorithms. What does "fair" mean in this context? Is fairness a mathematical property, a social value, or both?

□ Victoria Park Academy Shenzhen | IB DP Computer Science | Academic Year 2025–2026